

Hill Cipher

12. a. Write a C program to Encrypt the message “meet me at the usual place at ten rather than eightoclock” using the Hill cipher with the key.

9 4

5 7

a. Show your calculations and the result.

b. Show the calculations for the corresponding decryption of the ciphertext to recover the original plaintext.

12(a). Encryption

Aim

To encrypt the given plaintext using the Hill cipher with the key matrix.

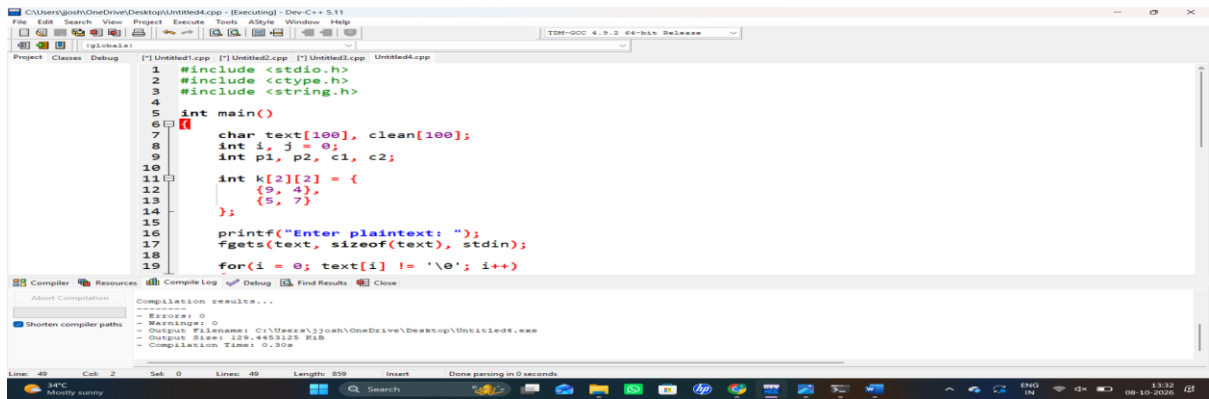
Key Matrix

$$[K = \begin{bmatrix} 9 & 4 \\ 5 & 7 \end{bmatrix}]$$

Algorithm

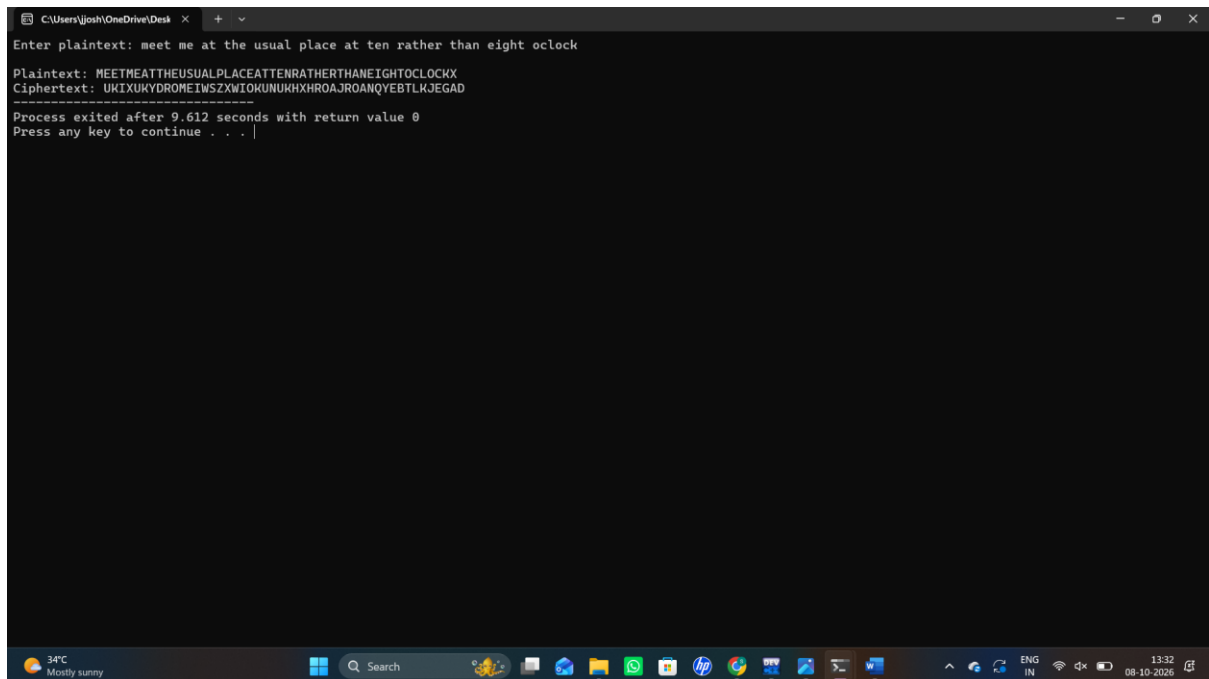
1. Convert the plaintext into uppercase.
2. Remove spaces.
3. Divide the plaintext into pairs.
4. Assign values A=0, B=1, ..., Z=25.
5. Multiply each plaintext pair by the key matrix.
6. Take modulo 26.
7. Convert the resulting numbers into letters.
8. Display the ciphertext.

Program



```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char text[100], clean[100];
7     int i, j = 0;
8     int p1, p2, c1, c2;
9
10    int k[2][2] = {
11        {9, 4},
12        {5, 7}
13    };
14
15    printf("Enter plaintext: ");
16    fgets(text, sizeof(text), stdin);
17    for(i = 0; text[i] != '\0'; i++)
```

Input/Output



```
Enter plaintext: meet me at the usual place at ten rather than eight o'clock
Plaintext: MEETMEATTHEUSUALPLACEATTENRATHERTHANEIGHTOLOCKX
Ciphertext: UKIXUKYDROMEIWSZXWIOKUNUKHXHROAJROANQYEBTLKJEGAD
-----
Process exited after 9.612 seconds with return value 0
Press any key to continue . . .
```

Result

Thus, the given plaintext was successfully encrypted using the Hill cipher.

12(b). Hill Cipher Decryption

Aim

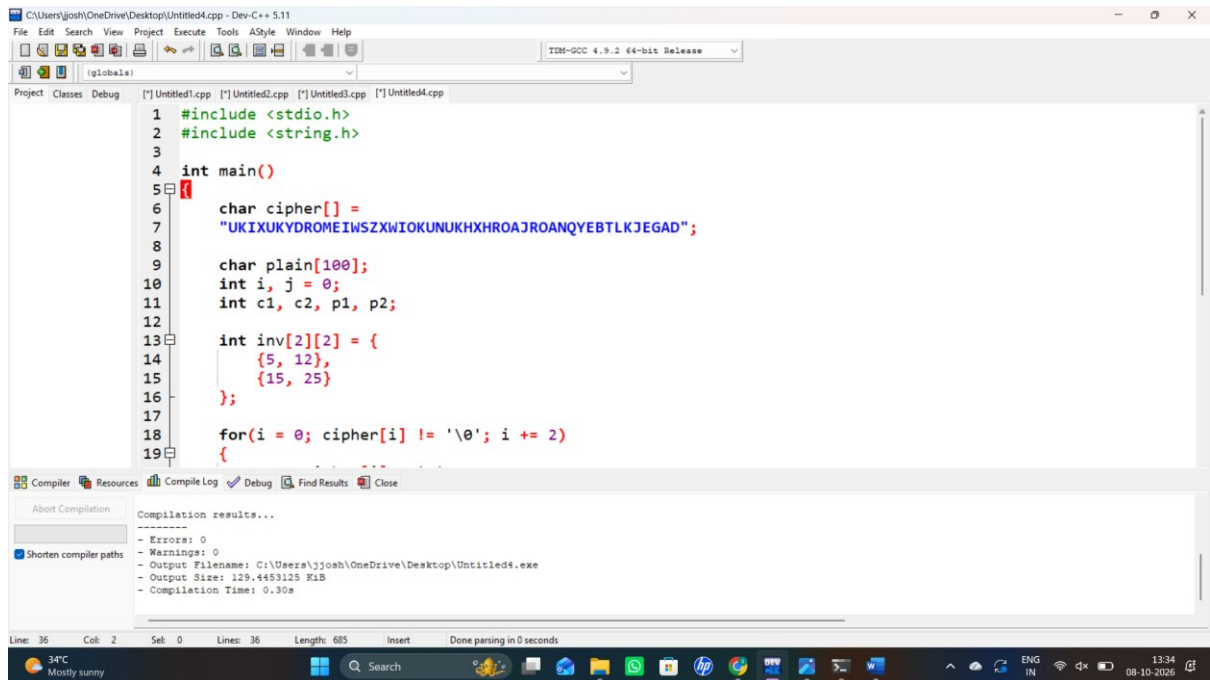
To decrypt the Hill cipher ciphertext and recover the original plaintext.

Algorithm

1. Find the determinant of the key matrix.
2. Calculate the determinant modulo 26.
3. Find its multiplicative inverse modulo 26.
4. Calculate the inverse key matrix.

5. Multiply each ciphertext pair by the inverse matrix.
6. Take modulo 26.
7. Convert the numbers into letters.
8. Remove the padding X.
9. Display the original plaintext.

Program

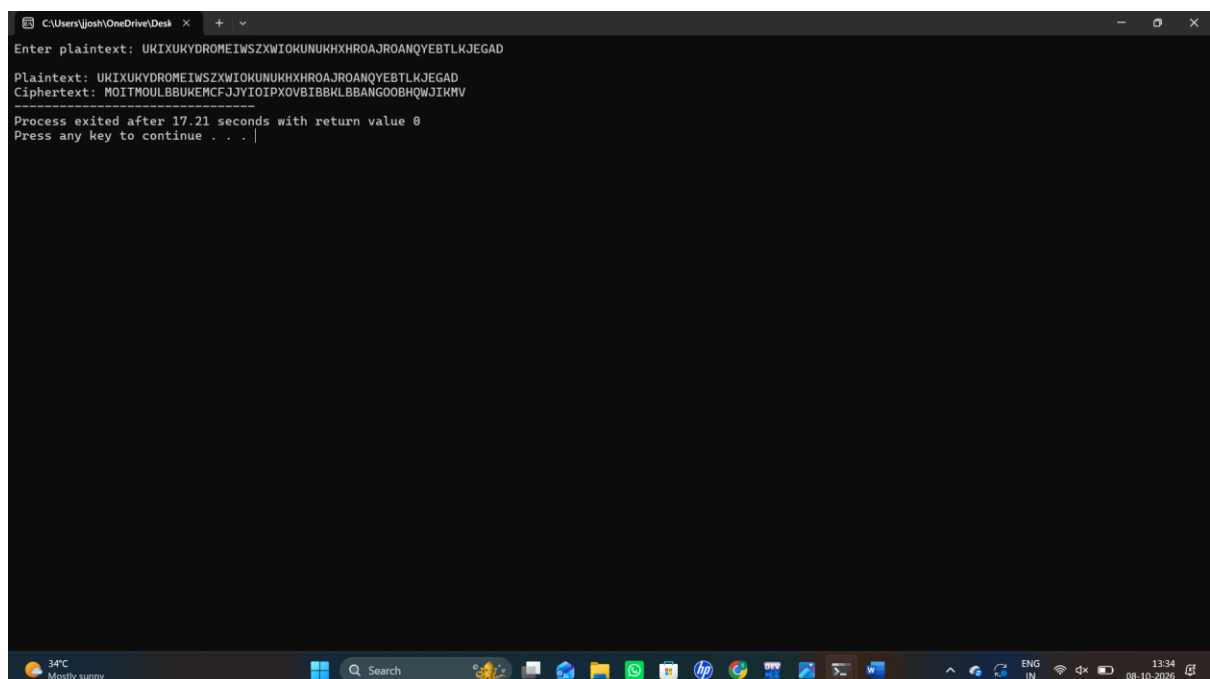


```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char cipher[] =
7     "UKIXUKYDROMEIWSZXWIKUNUKHXHROAJROANQYEBTLKJEGAD";
8
9     char plain[100];
10    int i, j = 0;
11    int c1, c2, p1, p2;
12
13    int inv[2][2] = {
14        {5, 12},
15        {15, 25}
16    };
17
18    for(i = 0; cipher[i] != '\0'; i += 2)
19    {
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\jjoah\OneDrive\Desktop\Untitled4.exe
- Output Size: 129.4453125 Kib
- Compilation Time: 0.30s

Input/Output



```
Enter plaintext: UKIXUKYDROMEIWSZXWIKUNUKHXHROAJROANQYEBTLKJEGAD
Plaintext: UKIXUKYDROMEIWSZXWIKUNUKHXHROAJROANQYEBTLKJEGAD
Ciphertext: MOITMOULBBUKEMCFJJYIIPXOVIBBKLBBANGOOBHQWJIKMV
Process exited after 17.21 seconds with return value 0
Press any key to continue . . .
```

Result

Thus, the ciphertext was successfully decrypted and the original plaintext was recovered.